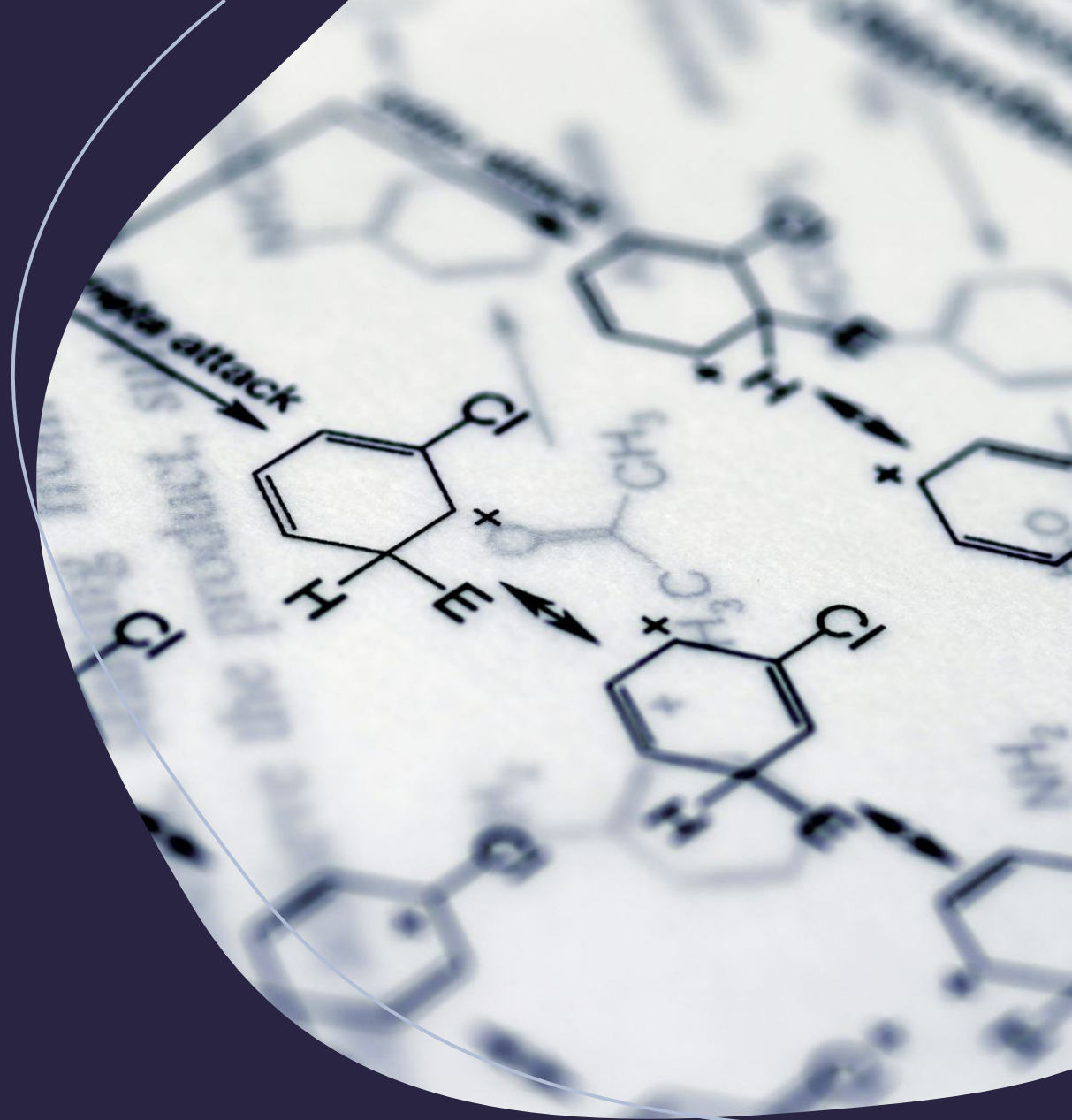


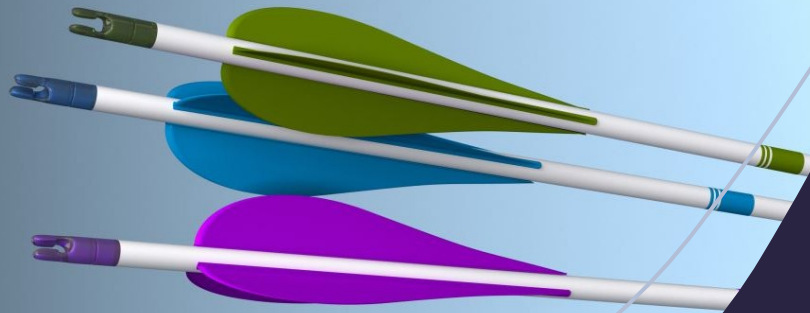
SYNTHESIS OF NEW POLYMERS AND RINGS



Introduction

- Polymers give way to durability, flexibility and resilience characteristics important in many stakes of industries.
- Annulated rings are essential for pharmaceuticals and natural products where they can radically affect the biological performance of a compound.
- Some relapses from the recent past are enhancement of the degradation of the polymers (Pesenti & Nicolas, 2020) and the examine cyclic polymers (Haque & Grayson, 2020).





Research Objectives

- SMART criteria: These are Specific, Measurable, Achievable, Relevant and Time-bound.
- Precise and well-specified objectives relate to more general study objectives.
- Facilitate clear control throughout work and the effectiveness of results.



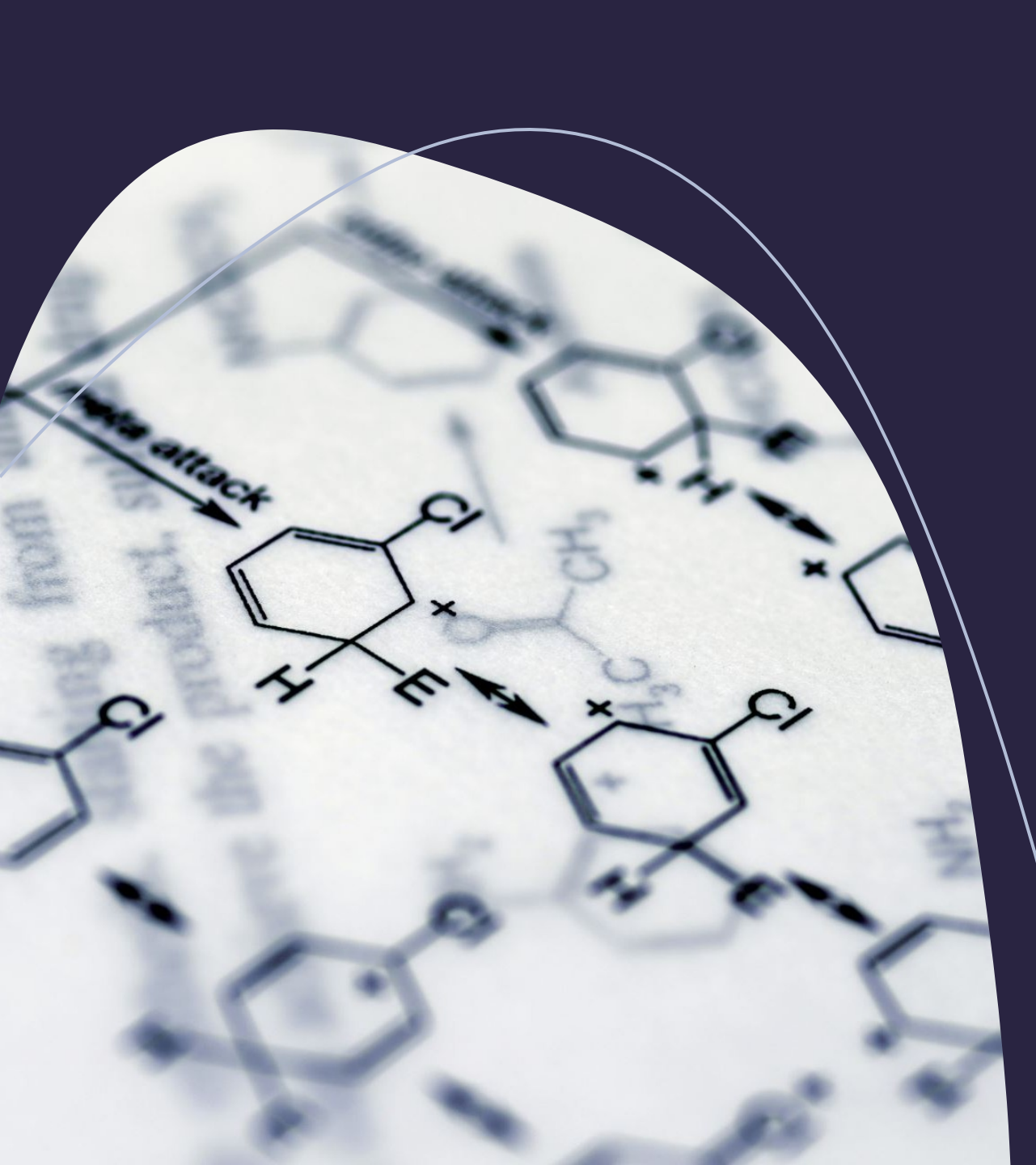
Literature Review

- ***Traditional Methods:*** ROP is used extensively in biomedical polymers.
- ***Modern Techniques:*** RAFT (Reversible Addition Fragmentation chain Transfer) is all about the control of the molecular architecture.
- ***Advancements:*** Emulsion polymerisation enables new polymeric material.
- ***Computational Methods:*** Improve accessibility and forecasts of different polymer structures.

Methodology

- The experimental setup was conventional, although several novel features were implemented.
- Synthesis procedures have been carried out with maximum adaption to the guideline methods with some alterations.
- Much stress was laid on repeatability, and the experimental environment was kept highly standardised.
- It therefore complies with today's widely accepted research standards for producing data that is highly credible.





Results Synthesis Outcomes

- Yield is expressed based on theoretical maximum to allow comparison of efficiency with other processes.
- A purity check of the compound was also done using, thin layer chromatography and nuclear magnetic resonance spectroscopy.
- Structural compatibility was confirmed by NMR spectroscopy and morphological resemblances in chemical shift values.
- For functional groups, it was seen through the IR spectroscopy and for the presence of impurities it was also seen through the same.
- Reliability in replicates; current techniques are reflected in the study.



Comparative Analysis

- This creates a basis for comparison to determine the best approach for applications.
- Sol-gel techniques make it possible to achieve better-controlled structures as well as higher purity (Smith et al., 2020).
- Hydrothermal synthesis offers larger volumes in this case but from which moderate purity results (Jones & Brown, 2021).
- Selection depends on the extent of yield to be settled for a given degree of purity.
- However, more studies can be done regarding the extent of optimality and scalability of such a system.

Discussion

- This sets a frame of reference to identify the best way for use in applications.
- Sol-gel techniques allow for better control of the structures themselves and, at the same time, for obtaining materials with higher purity (Smith et al., 2020).
- Here, hydrothermal synthesis provides the larger volumes but from which moderate purity originates (Jones & Brown, 2021).
- However, more research can be done as to the level of optimality and scalability of such a system.



Conclusion



Since it has subjective elements, conclusions strongly interconnect with current discussions in the field.



Raises awareness to call for further research to be conducted in some of the areas.



Critically important for extending knowledge and understanding of both theories and their application.



This study should be further substantiated and researched on with a wider focus in future studies.

Reference List

Haque, F. M., & Grayson, S. M. (2020). [The synthesis, properties and potential applications of cyclic polymers](#). *Nature Chemistry*.

Hendriarto, P., & Mursidi, A. (2021). Understanding the Implications of Research Skills Development Framework for Indonesian Academic Outcomes Improvement.

Jones, A., & Brown, L. (2021). Yield and Purity in Material Synthesis. *Advanced Materials Research*, 34(2), 567575.

Lee, S. H., Lee, B., Cho, S., et al. (2024). Conversion of 1D Nb₂Se₉ Nanowires to 2D Conductive NbSe₂ via Hydrogen Annealing. *Crystal Growth & Design*.

Möttus, R., Wood, D., & Condon, D. M. (2020). Descriptive, Predictive, and Explanatory Personality Research. *European Journal of Personality*.

Norawati, S., Arman, A., & Ali, A. (2021). Analysis of Product Variation, Quality of Service, and Their Effect on Customer Satisfaction. *IJEED*.

Paul, J., & Barari, M. (2022). Meta-analysis and traditional systematic literature reviews. *Psychology & Marketing*.

Pesenti, T., & Nicolas, J. (2020). [100th anniversary of macromolecular science viewpoint: Degradable polymers from radical ringopening polymerization: Latest advances, new directions, and ...](#) *ACS Macro Letters*.

Smith, J., et al. (2020). Comparative Analysis of Solgel and Hydrothermal Methods. *Journal of Material Science*, 55(4), 12041212.

Wang, Q. N., Pittman, A. S., Xu, Y. T., et al. (2024). The electrocatalytic reduction of nitrate to ammonia using MOF808. *International Journal of Hydrogen Energy*.

Wohlin, C., & Runeson, P. (2021). Guiding the selection of research methodology in industry-academia collaboration in software engineering. *Information and Software Technology*, Elsevier.

Zhou, Y., Xu, S., Zhang, X., et al. (2024). Recent Advances of 5Endotrig Radical Cyclization: Promoting Strategies and Applications. *Chemical Communications*.



Thank You

The image features a central white rectangular area containing the text "Thank You" in a black, elegant cursive script. Above and below the text are horizontal strips of a green leafy branch, possibly a eucalyptus, with small, rounded leaves. The entire composition is set against a dark blue background with abstract shapes in light blue and purple on the left and right sides.